

New Generation English B· Diagnostic Test

C2 (Proficiency Diagnostic) B· Full Test

READING

Why simplicity in scientific theory is sometimes misleading

Few ideas in science enjoy as much rhetorical favour as simplicity. Researchers invoke Occam's razor; teachers cite Newton's elegant equations; popular writers admire those theories that can be condensed into a single sentence. There is, however, a quieter scientific tradition that holds simplicity to be a useful working preference rather than a reliable indicator of truth. According to this view, the universe has no obligation to be elegant, and the conviction that it must be can become its own intellectual hazard.

The argument is not that simplicity is unimportant. A theory with fewer arbitrary parameters is easier to test, easier to teach and harder to disguise as something it is not. These are genuine virtues. The problem appears when simplicity becomes a heuristic so deeply assumed that it operates as a hidden axiom when researchers reach instinctively for the simpler model because they believe nature must agree, rather than because the evidence has yet decided between competing alternatives.

Historical examples are instructive. Early models of planetary motion before Kepler were aesthetically attractive but explanatorily poor. Pre-genetic models of inheritance were similarly tidy. In both cases, the simpler description was, for a while, mistaken for the more correct description. Acceptance of the messier alternative required the field to tolerate temporary inelegance to live, intellectually, in a less polished house until the new theory could be properly furnished.

This lesson has methodological consequences. A research culture that punishes complexity as inelegant will systematically slow the discovery of phenomena whose actual structure happens to be complex. Biology is the obvious example. So is climate science, where the genuine answer involves dozens of variables behaving non-linearly, and where each attempt to reduce the picture to a clean slogan misrepresents the science it intends to summarise. Strangely, the strongest demand for simplification often comes from outside the discipline rather than from within it.

The wisest formulation, perhaps, is to treat simplicity as a default rather than a doctrine. Begin with simple models, prefer them when they fit, and abandon them without sentiment when the evidence does not cooperate. Refuse, however, to mistake parsimony for proof. A theory survives because the world appears to behave the way it predicts, not because it pleases our taste for tidiness. To remember this and to insist on it inside disciplines that are quietly drifting away from it is one of the small, ungrateful jobs that careful thinkers continue to do, often without applause.

1. What principle of simplicity does the writer mention by name?

- A) Pascal's wager.
- B) Occam's razor.**
- C) G del's theorem.

2. According to the "quieter scientific tradition", how should simplicity be regarded?

- A) As a useful working preference rather than a reliable indicator of truth.**
- B) As the central law of nature.

C) As an emotional reaction unworthy of science.

3. What are the genuine virtues of a simpler theory?

A) Easier to test, easier to teach, harder to disguise as something it is not.

B) Always more correct than complex alternatives.

C) Cheaper to publish.

4. When does simplicity become a problem, according to the writer?

A) When it operates as a hidden axiom "вѣ" assumed before the evidence has decided.

B) When it is openly discussed.

C) When it is taught in universities.

5. Which two historical fields are given as examples of false simplicity?

A) Mathematics and rhetoric.

B) Planetary motion before Kepler, and pre-genetic models of inheritance.

C) Optics and acoustics.

6. What does the writer say about biology and climate science?

A) They are obvious examples of phenomena whose actual structure is complex.

B) They are perfectly simple subjects.

C) They have already been fully understood.

7. Where does the strongest demand for simplification often come from, according to the writer?

A) From inside the discipline.

B) From outside the discipline rather than from within it.

C) From foreign governments.

8. What is the writer's "wisest formulation"?

A) Treat simplicity as a default rather than a doctrine.

B) Always insist on the simplest possible theory.

C) Ban complex models entirely.

LISTENING

Seminar response: pushing back on the memory-history opposition

Script:

Let me push back briefly on a claim made in your last paper. You wrote that memory and history are opposed, and that the historian's task is to correct memory with evidence. I think this framing, while attractive, conceals more than it reveals.

Consider the following: in many historical contexts, communal memory has preserved patterns that documentary evidence missed entirely. Stories of pre-industrial epidemics, of unrecorded labour movements, of localised famines — these survived for generations in oral testimony before written sources caught up. Sometimes the documents were never produced, and sometimes they were produced selectively by people with their own interests. Memory, in those cases, was not the failed shadow of history; it was the only available record.

This does not mean that memory is reliable in the way documents can be. It is selective, suggestible and reshaped by present-day concerns. But the documentary record is also selective. The question, in any concrete case, is not which source is purer but which gives us the more honest reconstruction when combined with the other.

The danger of a strong opposition between memory and history is that it can seem to authorise the dismissal of communal recollection whenever it conflicts with paperwork. In practice, this dismissal has often served the powerful, because paperwork is more likely to survive when produced by the powerful. A history that systematically privileges documents will systematically under-record what was important to people who could not write, were not allowed to write, or wrote in forms we have not preserved.

So I would suggest a more dialectical formulation. History is what survives critical examination; memory is what a community lives by. The serious historian does not choose between them. The serious historian uses each to interrogate the other — and accepts, sometimes uncomfortably, that the most accurate account is produced not by adjudicating between them but by reading them together.

1. What is the speaker pushing back on?

A) A claim that memory and history are opposed and that historians correct memory.

B) A new historical method.

C) A particular textbook.

2. What kinds of episodes survived in oral testimony before written sources caught up?

A) Wars only.

B) Pre-industrial epidemics, unrecorded labour movements, localised famines.

C) Royal weddings.

3. Why were some documents never produced or selectively produced?

A) Because of bad weather.

B) Because of the interests of those who could produce them.

C) Because of religious bans.

4. Does the speaker claim memory is reliable in the way documents can be?

A) Yes, always.

B) No — it is selective, suggestible and reshaped by present concerns.

C) Memory is the only reliable source.

5. What is also true of the documentary record, according to the speaker?

A) It is fully objective.

B) It is also selective.

C) It is mostly invented.

6. What is the danger of strong opposition between memory and history?

A) It can seem to authorise dismissing communal recollection.

B) It speeds up research.

C) It encourages translation.

7. Who has often benefited from privileging documents over memory?

A) The poor.

B) The powerful.

C) The young.

8. What is the speaker's preferred formulation?

A) History replaces memory.

B) Memory replaces history.

C) A dialectical reading where each interrogates the other.

WRITING

Write 350-450 words. A respected historian recently argued that collective memory is closer to mythology than to fact, and that one of the historian's central duties is to correct it. Other voices reply that documentary evidence is itself shaped by power and that memory often preserves what records suppress. Discuss both views, and explain to what extent you find the distinction between memory and history useful. Qualify your position carefully.

SPEAKING

Record 4 minutes as if responding in an academic seminar. Engage with the question: do historians have a duty to challenge collective memory? Construct an argument with one strong concession, one specific historical case, and a final refined position that avoids vague balance.

GRAMMAR (RANGE & ACCURACY)

1. It is imperative that researchers ___ simpler models without sentiment when evidence does not cooperate.

A) abandon въ“ B) abandons C) are abandoning

2. Only when evidence converges ___ simplicity be accepted as proof.

A) can въ" B) it can C) should

3. Choose the simpler model instinctively, scientists sometimes overlook the truth.

A) Preferring **ВЪ**“ B) Preferred C) Having preferred

4. The universe ____ have no obligation to be elegant.

A) may B) can C) shall

5. the field tolerated complexity earlier, progress would have been faster.

A) Had бы" B) If C) Should

6. Theories that as inelegant are sometimes the more correct ones.

A) dismissing B) dismissed C) having dismissed

7. What scientists must remember ____ that parsimony is not proof.

A) be B) is B) is C) it is

8. that simplicity is a default, it should not become a doctrine.

A) Given B) Considering C) Although

9. A research culture may ____ slow the discovery of complex phenomena.

A) well въ" B) much C) very

10. Pre-genetic models of inheritance were tidy; ____, they were also wrong.

A) rather B) however C) conversely

11. The strongest demand for simplification often comes ____ outside the discipline.

A) from B) of C) for

12. A theory survives because the world behaves the way it predicts, ____ because it pleases our taste for tidiness.

A) not вѣ“ B) no C) never